

二、PCIe 使用前检查

PCIe最常见的问题就是第一次使用时无法正常工作。

所以在使用PCIe之前，请根据本文档检查下面的电路图和软件配置是否符合要求，以确保PCIe可以正常工作。

2.1. 电路图

首先我们看电路图部分，一般分为PCIe RC插槽标准接口或者直接连接PCIe EP设备

2.1.1 EVB使用PCIe插槽

EVB使用PCIe标准插槽，可以灵活更换PCIe EP设备进行调试。如果您的EVB使用PCIe标准插槽，请确认以下几点。

- 1: 确认RC、EP的PERST/WAKE#/CLKREQ电压是否匹配，若不一致需要增加level-shift电路转换电压。
- 2: 确认WAKE#、CLKREQ、PERST是否接10K上拉电阻。（78电平转换内部有10k上拉）
- 3: 确认TXP、TXN是否接电容，PCIe GEN3、GEN4电容值为220nf，PCIe GEN1、GEN2电容值为100nf。
- 4: 确认PCIE_CKP/PCIE_CKN是否接49.9ohm下拉电阻。（78内部有49.9欧下拉，且不可取消）
- 5: 确认PCIe设备是否需要插槽12V、3.3V供电，以及插槽的供电线路是否正确。
- 6: 确认插槽信号损耗，部分插槽无法支持PCIe GEN3/GEN4。（89272只支持gen2）

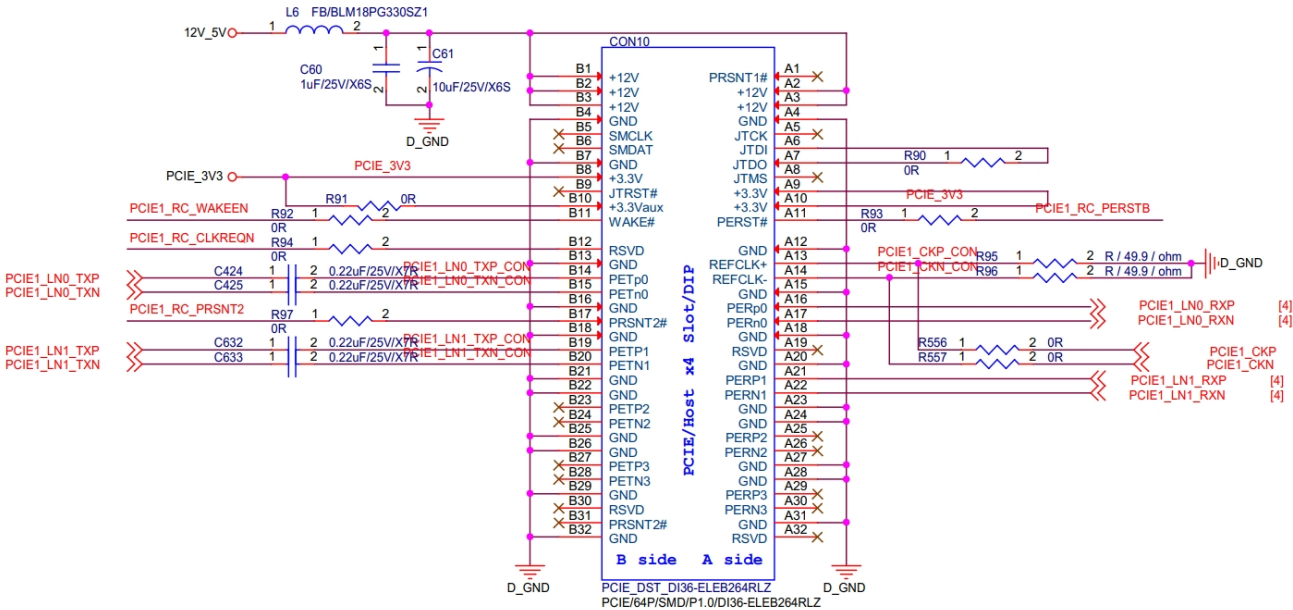


图 2-1. PCIe 插槽电路图

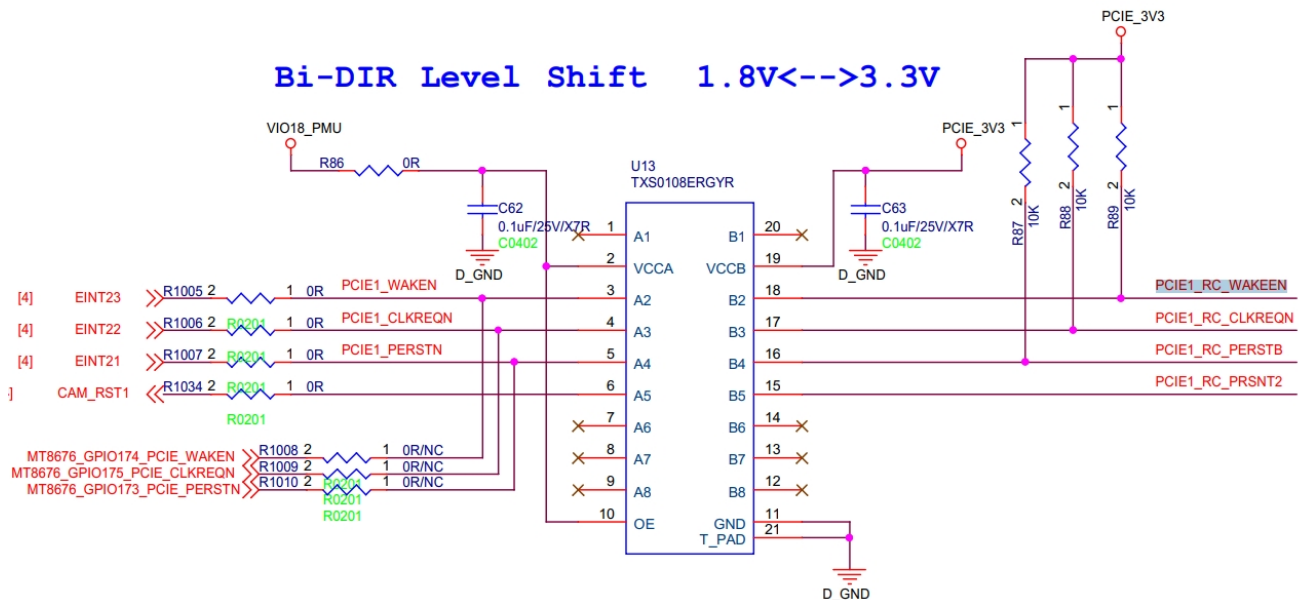


图 2-2. 电平转换电路图

2.1.2 直接连接PCIe EP（使用的这种方式）

直接连接EVB上的PCIe EP可以减少EVB的面积，如果您的EVB采用直连方式，请确认以下几点：

- 1：确认RC和EP的PERST/WAKE#/CLKREQ电压是否一致，如果不一致需要增加level-shift电路转换电压。
- 2：确认RC和EP WAKE#、CLKREQ、PERST连接是否正确。
- 3：确认WAKE#、CLKREQ、PERST是否连接10K上拉电阻。
- 4：确认TXP和TXN是否连接电容，PCIe GEN3和GEN4电容值为220nf，PCIe GEN1和GEN2电容值为100nf。
- 5：确认 PCIE_CKP/PCIE_CKN 是否接有 49.9ohm 下拉电阻。
- 6：确认 TX/RX 是否有额外接电阻或电容影响信号质量。
- 7：确认 PCIe 设备供电线路，确保对接前设备上电

2.2 软件配置检查

PCIe相关软件是否配置正确，下面以mt6990为例。

2.2.1 配置

确保项目配置中的 PCIE_MEDIATEK_GEN3=y 和 CONFIG_DEVICE_MODULES_PHY_MTK_PCIE=y

```

Symbol: PCIE_MEDIATEK_GEN3 [=y]
Type : tristate
Prompt: MediaTek Gen3 PCIe controller
Location:
-> Device Drivers
-> PCI support (PCI [=y])
(3) -> PCI controller drivers
Defined at drivers/pci/controller/Kconfig:244
Depends on: PCI [=y] && (ARCH_MEDIATEK [=y] || COMPILE_TEST [=n]) && PCI_MSI_IRQ_DOMAIN [=y]

```

Figure 2-3. CONFIG PCIE_MEDIATEK_GEN3

```

Symbol: PHY_MTK_PCIE [=y]
Type : tristate
Prompt: MediaTek PCIe-PHY Driver
Location:
-> Device Drivers
(1) -> PHY Subsystem
Defined at drivers/phy/mediatek/Kconfig:36
Depends on: ARCH_MEDIATEK [=y] && OF [=y]
Selects: GENERIC_PHY [=y]

```

Figure 2-4. CONFIG PHY_MTK_PCIE

2.2.2 DTS

chip.dtsi(ex:mt6990.dtsi)中的PCIe和PHY节点包含与PCIe相关的基本信息。确保它们与默认值一致。

```

4158     pciephy0: phy0@11310000 {
4159         compatible = "mediatek,mt6980-pcie-phy";
4160         #address-cells = <2>;
4161         #size-cells = <2>;
4162         #phy-cells = <0>;
4163         reg = <0 0x11310000 0 0x10000>;
4164         reg-names = "sif";
4165         phy-id = <0>;
4166         status = "disabled";
4167
4168         power-domains = <&scpsys MT6990_POWER_DOMAIN_PCIE0>;
4169     };

```

Figure 2-5. PCIe PHY DTSI node

```

+ mt6990.dtsi
4042     pcie0: pcie@11280000 {
4043         compatible = "mediatek,mt6990-pcie";
4044         reg = <0 0x11280000 0 0x2000>;
4045         reg-names = "pcie-mac";
4046         linux,pci-domain = <0>;
4047         #address-cells = <3>;
4048         #size-cells = <2>;
4049         interrupts = <GIC_SPI 288 IRQ_TYPE_LEVEL_HIGH>;
4050         bus-range = <0x00 0xff>;
4051         ranges = <0x82000000 0 0x30000000
4052                 0x0 0x30000000 0 0x10000000>;
4053         phys = <&pciephy0>;
4054         phy-names = "pcie-phy";
4055         status = "disabled";
4056
4057         clocks = <&pericfg_ao_clk CLK_PERAO_PERI2PCIE0__TX>,
4058                 <&vlpcfg_clk CLK_VLPCFG_PCIE_TL_32K>,
4059                 <&vlpcfg_clk CLK_VLPCFG_PCIE_TL_26M>,
4060                 <&vlpcfg_clk CLK_VLPCFG_PCIE_TL_96M>,
4061                 <&vlpcfg_clk CLK_VLPCFG_PCIE_TL_ULPOSC>,
4062                 <&vlpcfg_clk CLK_VLPCFG_PCIE_REF_CK_26M>,
4063                 <&vlpcfg_clk CLK_VLPCFG_PCIE_AXI250>,
4064                 <&vlpcfg_clk CLK_VLPCFG_PCIE_SRAM_26M>,
4065                 <&vlp_cksys_clk CLK_VLP_CK_TL_VLP_SEL>;
4066
4067         power-domains = <&scpsys MT6990_POWER_DOMAIN_PCIE0>;
4068
4069         #interrupt-cells = <1>;
4070         interrupt-map-mask = <0 0 0 7>;
4071         interrupt-map = <0 0 0 1 &pcie_intc0 0>,
4072                        <0 0 0 2 &pcie_intc0 1>,
4073                        <0 0 0 3 &pcie_intc0 2>,
4074                        <0 0 0 4 &pcie_intc0 3>;
4075         pcie_intc0: interrupt-controller {
4076             interrupt-controller;
4077             #address-cells = <0>;
4078             #interrupt-cells = <1>;
4079         };
4080     };

```

Figure 2-6. PCIe MAC DTSI node

project.dts(ex: evb6990_cpe.dts) 包含与项目相关的板级信息。

- 1: 确认要使用的 PCIe 和 PHY 节点处于启用状态 (status = "okay") 。
- 2: 确认 PCIe 节点中包含的 GPIO 与 EVB 使用的 GPIO 一致。
- 3: 确认 PCIe 节点中包含的 GPIO 状态是否正确切换到 PCIe 模式。

```

evb6990_cpe.dts
77 &pcie0 {
78     pinctrl-names = "default", "idle", "sleep";
79     pinctrl-0 = <&pcie0_pins_default>;
80     pinctrl-1 = <&pcie0_pins_idle>;
81     pinctrl-2 = <&pcie0_pins_sleep>;
82     status = "okay";
83 };
84

```

Figure 2-7. PCIe MAC DTS node

```

101 &pciephy0 {
102     status = "okay";
103 };

```

Figure 2-8. PCIe PHY DTS node

```

113 /* PCIe RC mode */
114 &pio {
115     pcie0_pins_default: pcie0-default {
116         mux {
117             pinmux = <PINMUX_GPIO54_FUNC_PCIE_PERESET_N_OP>,
118                     <PINMUX_GPIO55_FUNC_PCIE_WAKE_N_OP>,
119                     <PINMUX_GPIO56_FUNC_PCIE_CLKREQ_N_OP>;
120             drive-strength = <1>;
121             bias-pull-up;
122         };
123     };
124     pcie0_pins_idle: pcie0-idle {
125         mux {
126             pinmux = <PINMUX_GPIO54_FUNC_GPIO54>,
127                     <PINMUX_GPIO55_FUNC_GPIO55>,
128                     <PINMUX_GPIO56_FUNC_GPIO56>;
129             bias-pull-down;
130         };
131     };
132 };
133     pcie0_pins_sleep: pcie0-sleep {
134         mux {
135             pinmux = <PINMUX_GPIO54_FUNC_GPIO54>,
136                     <PINMUX_GPIO55_FUNC_GPIO55>,
137                     <PINMUX_GPIO56_FUNC_GPIO56>;
138             bias-pull-down;
139         };
140     };
141 };

```

Figure 2-9. PCIe GPIO node

三、PCIe驱动探测异常

完成 PCIe 预使用检查后，如果仍然无法正常识别 PCIe EP 设备，一般是驱动探测过程中 Linkup Fail 导致的，需要根据故障时的 LTSSM (Link Training and Status State Machine) 状态判断问题原因

3.1 检测状态下链接失败

PCIe 检测状态是通过电路上电平转换的速率来判断对端设备是否存在。MTK 平台日志“ltssm reg val:0x1”(图 3-1)表示链路状态处于检测阶段，无法识别对端设备。

此问题常见原因有：

- 1: PCIe RC 驱动探测前 PCIe EP 设备未上电。
- 2: PCIe EP 设备复位或 PERST (电平转换) 未解除。
- 3: PCIe EP 设备异常或 Linkup 时需要特殊时序。
- 4: TX/RX 差分信号连接错误。
- 5: TX/RX 线上没有电容或电容损坏

```

mtk-pcie 11280000.pcie: host bridge /pcie@11280000 ranges:
mtk-pcie 11280000.pcie: MEM 0x30000000..0x3fffffff -> 0x30000000
mtk-pcie 11280000.pcie: 11280000.pcie supply pcie1v8 not found, using dummy regulator
mtk-pcie 11280000.pcie: 11280000.pcie supply pcie3v3 not found, using dummy regulator
mtk-pcie 11280000.pcie: 11280000.pcie supply pcie12v not found, using dummy regulator
current sw version: E1
mtk-pcie 11280000.pcie: PCIe link down, ltssm reg val: 0x1
mtk-pcie: probe of 11280000.pcie failed with error -110

```

Figure 3-1. Linkup failure in detect state

3.2 polling状态下Linkup失败

PCIe轮询状态是RC和EP通过TS1/TS2握手获取信息、Bit lock和symbol lock。MTK平台日志“ltssm reg val:0x30000003”(图3-2)表示链路状态处于轮询阶段，RC和EP无法正常通信。

该问题常见原因有：

- 1：CLK差分信号没有接49.9ohm的地电阻，会导致没有100M时钟输出到EP设备。
- 2：TX/RX差分信号上电容异常导致信号质量差，无法处理正常握手。
- 3：PCIe EP设备异常或者有特殊要求，导致EP设备不回复TS1/TS2握手包
- 4：PCIe GPIO接了level-shift，level-shift异常或者没上电，导致时序异常。

```
mtk-pcie 11280000.pcie: host bridge /pcie@11280000 ranges:
mtk-pcie 11280000.pcie: MEM 0x30000000..0x3fffffff -> 0x30000000
mtk-pcie 11280000.pcie: 11280000.pcie supply pcielv8 not found, using dummy regulator
mtk-pcie 11280000.pcie: 11280000.pcie supply pcie3v3 not found, using dummy regulator
mtk-pcie 11280000.pcie: 11280000.pcie supply pcie12v not found, using dummy regulator
current sw version: E1
mtk-pcie 11280000.pcie: PCIe link down, ltssm reg val: 0x30000003
mtk-pcie: probe of 11280000.pcie failed with error -110
```

Figure 3-2. Linkup failure in polling state

3.3 其他状态下Linkup失败

Linkup fail一般出现在detect和polling状态下，如果在log中看到其他状态下Linkup failed，请提供对应log给MTK专家进行分析。

四、PCIe 暂停suspend/恢复resume异常

挂起/恢复功能是系统进入空闲模式时省电的重要功能。在挂起过程中，PCIe 进入 L2 状态，关闭电源和时钟。在恢复过程中，PCIe 打开电源和时钟并重新建立链接。当系统处于挂起状态时，PCIe 设备可以通过拉低 wake# 来唤醒系统。

4.1 PCIe suspend 异常

当 PCIe suspend 失败时，你会看到如下的日志（图 4-1）。最常见的原因是由于无法进入 L2 导致 suspend 失败。无法进入 L2 的常见原因有：

- 1：PCIe sideband 的 wake# 信号被 assert，导致 PCIe 无法进入 L2 状态。
- 2：PCIe 准备进入 L2 状态时，EP 设备已经断电，导致 PCIe 无法进入 L2 状态。
- 3：level-shift 或其他问题导致 PERST# 信号被 assert，导致 PCIe 无法进入 L2 状态。

```
mtk-pcie 11280000.pcie: cannot enter L2 state
PM: dpm_run_callback(): genpd_suspend_noirq+0x0/0x30 returns -110
PM: Device 11280000.pcie failed to suspend noirq: error -110
```

Figure 4-1. PCIe Suspend fail log

4.2 PCIe 恢复 (resume) 异常

当 PCIe resume 失败时，您将看到以下日志（图 4-2）。最常见的原因是由于 linkup fail 导致 resume 失败。有关如何调试 linkup fail，请参阅第 3 章。

```
mtk-soc-temp-lvts 14414000.lvts: [Thermal/LVTS]lvts_resume_noirq
mtk-pcie 16940000.pcie: PCIe link start ...
mtk-pcie 16940000.pcie: PCIe link down, ltssm reg val: 0x1
mtk-pcie 16940000.pcie: PM: dpm_run_callback(): genpd_resume_noirq+0x0/0x164 returns -110
mtk-pcie 16940000.pcie: PM: failed to resume async_noirq: error -110
```

Figure 4-2. PCIe Resume fail log

4.3 EP Assert wake# 无法唤醒系统

当系统处于挂起状态时，PCIe EP 可以通过拉低 wake# 来唤醒系统。EP 唤醒系统失败的常见原因如下

- 1: 确认靠近芯片的 wake# 引脚是否被拉低。可能是电平转换异常或 EP 无法将 wake# 拉低。
- 2: 确认 wake# 的 GPIO 引脚复用器是否切换到 PCIe 模式。
- 3: 使用 SPM 命令检查 PCIe 唤醒源是否已启用。

4.4 系统被 PCIe 异常唤醒

PCIe意外唤醒系统会造成功耗增加，可通过suspend唤醒日志确认唤醒源，PCIe唤醒如图4-3所示。

PCIe异常唤醒常见原因有：

- 1: wake#需要接上拉电阻，避免电压抖动唤醒系统
- 2: 平台支持多个PCIe RC，未使用的PCIe RC的wake#管脚应设置为nonPCIe模式。
- 3: 若没有EP唤醒系统的应用场景，可将平台的wake#设置为GPIO模式，避免意外唤醒。

```
suspend wake up by R12_PCIE_WAKE_B, timer_out = 80413, r13 = 0x1a40, debug_flag = 0x1003f3ff 0:
suspend wake up by R12_PCIE_WAKE_B, timer_out = 57447, r13 = 0x130c, debug_flag = 0x1003f3ff 0:
suspend wake up by R12_PCIE_WAKE_B, timer_out = 91295, r13 = 0x1a40, debug_flag = 0x1003f3ff (
suspend wake up by R12_PCIE_WAKE_B, timer_out = 81310, r13 = 0x1a40, debug_flag = 0x1003f3ff (
suspend wake up by R12_PCIE_WAKE_B, timer_out = 91093, r13 = 0x1200, debug_flag = 0x1003f3ff (
suspend wake up by R12_PCIE_WAKE_B, timer_out = 80299, r13 = 0x1200, debug_flag = 0x1003f3ff (
suspend wake up by R12_PCIE_WAKE_B, timer_out = 91429, r13 = 0x130c, debug_flag = 0x1003f3ff (
suspend wake up by R12_PCIE_WAKE_B, timer_out = 81231, r13 = 0x1200, debug_flag = 0x1003f3ff (
suspend wake up by R12_PCIE_WAKE_B, timer_out = 90514, r13 = 0x1200, debug_flag = 0x1003f3ff (
suspend wake up by R12_PCIE_WAKE_B, timer_out = 90514, r13 = 0x1200, debug_flag = 0x1003f3ff (
suspend wake up by R12_PCIE_WAKE_B, timer_out = 70935, r13 = 0x1a40, debug_flag = 0x1003f3ff (
suspend wake up by R12_PCIE_WAKE_B, timer_out = 81270, r13 = 0x1200, debug_flag = 0x1003f3ff (
```

Figure 4-3. System wakeup by PCIe

五、Q&A

5.1 如何开启 PCIe ASPM?

- 1: 首先在内核配置中开启 PCI Express ASPM 控制。
- 2: 然后根据需要开启的 L0S/L1/L1.1/L1.2 状态开启相应的配置。
- 3: 最后重新编译使配置生效。

CONFIG_PCIEASPM:

This enables OS control over PCI Express ASPM (Active State Power Management) and Clock Power Management. ASPM supports capable state L0/L0s/L1.

ASPM is initially set up by the firmware. With this option enabled, Linux can modify this state in order to disable ASPM on known-bad hardware or configurations and enable it when known-safe.

PCI Ex

ASPM can be disabled or enabled at runtime via
/sys/module/pcie_aspm/parameters/policy

Debug

When in doubt, say Y.

Symbol: PCIEASPM [=y]

Type : bool

Prompt: PCI Express ASPM control

Location:

-> Device Drivers

-> PCI support (PCI [=y])

-> PCI Express Port Bus support (PCIEPORTBUS [=y])

Defined at drivers/pci/pcie/Kconfig:64

Depends on: PCI [=y] & PCIEPORTBUS [=y]

Figure 5-1. CONFIG_PCIEASPM

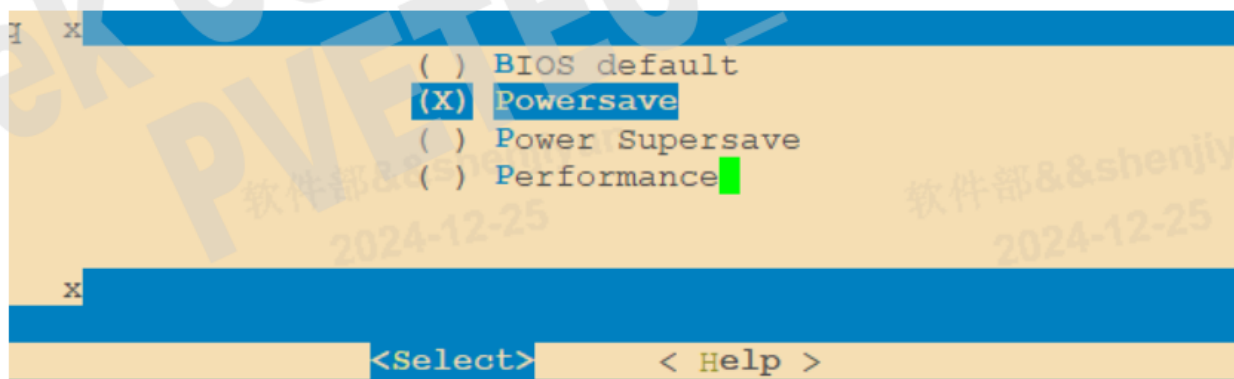


Figure 5-2. ASPM State

```
config PCIEASPM_POWERSAVE
    bool "Powersave"
    depends on PCIEASPM
    help
        Enable PCI Express ASPM L0s and L1 where possible, even if the
        BIOS did not.

config PCIEASPM_POWER_SUPERSAVE
    bool "Power Supersave"
    depends on PCIEASPM
    help
        Same as PCIEASPM_POWERSAVE, except it also enables L1 substates where
        possible. This would result in higher power savings while staying in L1
        where the components support it.

config PCIEASPM_PERFORMANCE
    bool "Performance"
    depends on PCIEASPM
    help
        Disable PCI Express ASPM L0s and L1, even if the BIOS enabled them.
```

Figure 5-3. ASPM state detail

5.2 如何开启 AER (Advanced Error Report)?

在内核配置中开启 CONFIG_PCIEAER 并重新编译使配置生效。

Enable CONFIG_PCIEAER in kernel config and recompile to make the config take effect.

```
CONFIG_PCIEAER:
This enables PCI Express Root Port Advanced Error Reporting
(AER) driver support. Error reporting messages sent to Root capable
Port will be handled by PCI Express AER driver.

Symbol: PCIEAER [=y]
Type : bool
Prompt: PCI Express Advanced Error Reporting support upport
Location: PCI Ex
-> Device Drivers
-> PCI support (PCI [=y]) control
-> PCI Express Port Bus support (PCIEPORTBUS [=y])
Defined at drivers/pci/pcie/Kconfig:25
Depends on: PCI [=y] && PCIEPORTBUS [=y]
Selects: RAS [=y]
```

Figure 5-4. CONFIG_PCIEAER

5.3 如何在 KO 中构建 PCIe 驱动程序?

在内核配置中设置 CONFIG_PCIE_MEDIATEK_GEN3 = m, 然后重新编译以使配置生效。

```
x
Cadence PCIe controllers support --->
[ ] Faraday Technology FTPCI100 PCI controller
[ ] Generic PCI host controller
[ ] Xilinx AXI PCIe host bridge support
[ ] X-Gene PCIe controller
< > Altera PCIe controller
[ ] Cavium Thunder PCIe controller to off-chip devices
[ ] Cavium Thunder ECAM controller to on-chip devices on pass-1.x silicon
< > MediaTek PCIe controller
<M> MediaTek Gen3 PCIe controller
DesignWare PCI Core Support --->
```

Figure 5-5. CONFIG_PCIE_MEDIATEK_GEN3

5.4 如何在用户空间读写寄存器?

在内核和busybox配置中启用CONFIG_DEVMEM, 然后重新编译以使配置生效。

```
Symbol: DEVMEM [=n]
Type : bool
Prompt: /dev/mem virtual device support
Location:
-> Device Drivers
(1) -> Character devices
Defined at drivers/char/Kconfig:10
```

Figure 5-6. CONFIG_DEVMEM

then use follow command get how to use it.

```
k6991v1_64:/ # devmem --help
Toybox 0.8.11-android multical1 binary (see toybox --help)

usage: devmem ADDR [WIDTH [DATA]]

Read/write physical address. WIDTH is 1, 2, 4, or 8 bytes (default 4).
Prefix ADDR with 0x for hexadecimal, output is in same base as address.
```

Figure 5-7. Command 'devmem --help'

5.5 如何限制 PCIe 速度?

通过修改DTS的PCIe节点中的'max-link-speed', 可以限制PCIe的最大速度。例如:

```
Limit to GEN4 ---> max-link-speed = <4>;
Limit to GEN3 ---> max-link-speed = <3>;
Limit to GEN2 ---> max-link-speed = <2>;
Limit to GEN1 ---> max-link-speed = <1>;
```



```
&pcie0 {
    pinctrl-names = "init", "default", "sleep", "pd";
    pinctrl-0 = <&pcie0_pins_init>;
    pinctrl-1 = <&pcie0_pins_default>;
    pinctrl-2 = <&pcie0_pins_sleep>;
    pinctrl-3 = <&pcie0_pins_pd>;
    max-link-speed = <4>;
    status = "okay";
};
```

Figure 5-8. The max-link-speed property

5.6 IO空间如何分配?

联发科默认把MMIO空间全部分配为MEM空间, 如下图5-9所示。

- 1: 0x81000000代表IO空间, 0x82000000代表MEM空间。
- 2: 第二、三个字段代表PCI域的基地址。
- 3: 第四、五个字段代表CPU域的基地址。
- 4: 最后两个字段代表映射区域的大小

```
interrupts = <GIC_SPI 288 IRQ_TYPE_LEVEL_HIGH>;
bus-range = <0x00 0xff>;
ranges = <0x82000000 0 0x30000000
0x0 0x30000000 0 0x100000000>;
status = "disabled";
```

Figure 5-9. Default MMIO Space

因此，如果你想为设备分配 IO 空间，你应该像图 5-10 那样更改范围

```
interrupts = <GIC_SPI 288 IRQ_TYPE_LEVEL_HIGH>;
bus-range = <0x00 0xff>;
ranges = <0x82000000 0 0x30000000
0x0 0x30000000 0 0x80000000>,
<0x81000000 0 0x38000000
0x0 0x38000000 0 0x80000000>;
status = "disabled";
```

Figure 5-10. Add IO space in range

5.7 如何唤醒有 PCIe 设备的系统？

当系统处于挂起状态时，PCIe 设备可以通过拉低 WAKE# 来唤醒系统

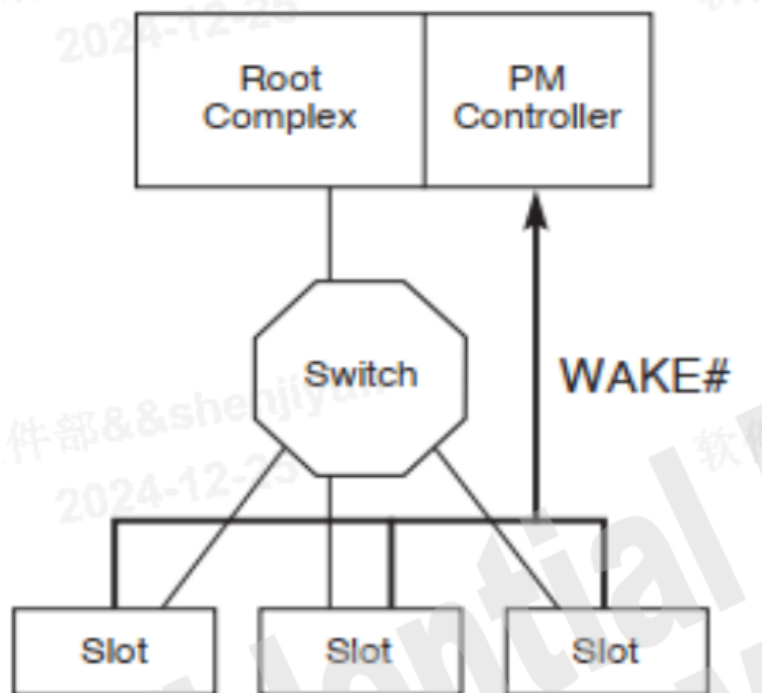


Figure 5-11. System wakeup by PCIe

5.8 如何查看 PCIe 基本信息？

lspci 工具是一个显示系统中 PCI 总线以及与其连接的设备信息的实用程序。使用 lspci 可以查看 PCIe 总线上所有设备的基本信息和能力配置信息。如下图 5-12 所示：

```
k6991v1_64:/data # ./lspci -vv
00:00.0 Class 0604: Device 14c3:68a0 (rev 01)
Control: I/O- Mem+ BusMaster+ SpecCycle- MemWINV- VGASnoop- ParErr- Stepping- SERR- FastB2B- DisINTx+
Status: Cap+ 66MHz- UDF- FastB2B- ParErr- DEVSEL=fast >TAbort- <TAbort- <MAbort- >SERR- <PERR- INTx-
Latency: 0
Interrupt: pin A routed to IRQ 664
Region 0: Memory at 50300000 (64-bit, non-prefetchable) [size=32K]
Bus: primary=00, secondary=01, subordinate=01, sec-latency=0
I/O behind bridge: [disabled]
Memory behind bridge: 50000000-502fffff [size=3M]
Prefetchable memory behind bridge: [disabled]
Secondary status: 66MHz- FastB2B- ParErr- DEVSEL=fast >TAbort- <TAbort- <MAbort- <SERR- <PERR-
BridgeCtl: Parity- SERR+ NoISA- VGA- VGA16- MAbort- >Reset- FastB2B-
PriDiscTmr- SecDiscTmr- DiscTmrStat- DiscTmrSERREN-
Capabilities: [80] Express (v2) Root Port (Slot-), MSI 00
DevCap: MaxPayload 512 bytes, PhantFunc 0
ExtTag+ RBE+
DevCtl: CorrErr+ NonFatalErr+ FatalErr+ UnsupReq+
RlxdOrd+ ExtTag+ PhantFunc- AuxPwr- NoSnoop+
MaxPayload 128 bytes, MaxReadReq 512 bytes
DevSta: CorrErr- NonFatalErr- FatalErr- UnsupReq- AuxPwr- TransPend-
LnkCap: Port #1, Speed 16GT/s, Width x1, ASPM L0s L1, Exit Latency L0s <2us, L1 <8us
ClockPM- Surprise+ LLActRep+ BwNot+ ASPMOptComp+
LnkCtl: ASPM L1 Enabled; RCB 64 bytes, Disabled- CommClk+
ExtSynch- ClockPM- AutWidDis- BWInt- AutBWInt-
LnkSta: Speed 16GT/s (ok), Width x1 (ok)
TrErr- Train- SlotClk+ DLActive+ BWMgmt- ABWMgmt-
RootCap: CRSVisible+
RootCtl: ErrCorrectable- ErrNon-Fatal- ErrFatal- PMEIntEna+ CRSVisible+
RootSta: PME ReqID 0000, PMEStatus- PMEPending-
DevCap2: Completion Timeout: Range ABCD, TimeoutDis+ NROPrPrP- LTR+
10BitTagComp+ 10BitTagReq- OBFF Not Supported, ExtFmt+ EETLPPrefix-
EmergencyPowerReduction Not Supported, EmergencyPowerReductionInit-
FRS- LN System CLS Not Supported, TPHComp- ExtTPHComp- ARIFwd-
AtomicOpsCap: Routing- 32bit- 64bit- 128bitCAS-
DevCtl2: Completion Timeout: lms to 10ms, TimeoutDis- LTR+ OBFF Disabled, ARIFwd-
AtomicOpsCtl: ReqEn- EgressB1ck-
LnkCap2: Supported Link Speeds: 2.5-16GT/s, Crosslink- Retimer+ 2Retimers+ DRS-
LnkCtl2: Target Link Speed: 16GT/s, EnterCompliance- SpeedDis-
Transmit Margin: Normal Operating Range, EnterModifiedCompliance- ComplianceSOS-
Compliance De-emphasis: -6dB
```

Figure 5-12. Lspci -vv result

关于lspci的详细信息，请参考以下网页

<https://man7.org/linux/man-pages/man8/lspci.8.html>